CALCULUS · UNIT 7: APPLICATIONS OF INTEGRALS

Glass Vase Studio

Use area between curves and volumes of revolution to design glass vases and bottles.

Honors

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

Lumen Glass Studio makes vases, bottles, and paperweights. Each design is a curve spun around an axis, the logo is a region between two curves, and glass is priced by volume.

You will use integrals to calculate the area and volume of each design and price the products.



Glass Price

$0.08 per cm³

Vase Profile

$r = 4 + \sin(\pi x/10)$

x in cm, $0 \le x \le 20$

Learning Goals

- I can set up and evaluate area and volume integrals about any axis.
- I can compare methods and use technology appropriately.
- I can optimize a design using integral models.

Project Timeline

When	What you will do
Day 1	Logo Design
Day 2	Vase Engineering
Day 3	Paperweights
Final	Product Design Portfolio + reflection

Part 1: Logo Design

Day 1 · Use the Design Specification Cards.

Design Specification Cards (units: cm)

Logo	Bud Vase	Tall Vase
Between $y = \sqrt{x}$	$y = \sqrt{x}$ rotated about	$r = 4 + \sin(\pi x/10)$
and $y = x/2$	the x-axis, $0 \le x \le 16$	$0 \le x \le 20$

Hollow Vase	Paperweight	Production
Outer $r = 4 + \sin(\pi x/10)$	Base: $y = 4 - x^2$, $y \ge 0$	Rate $p(t) = 6 + 2t$
inner $r = 3.5 + \sin(\pi x/10)$	square cross sections	vases per hour

1. Find the area between $y = \sqrt{x}$ and $y = \frac{x}{2}$ and the line $x = 9$ (two regions).

2. Find k so the vertical line $x = k$ splits the logo ($0 \le x \le 4$) into two equal areas.

3. Rotate the logo about the x-axis. Find the volume (washer method).

4. Rotate the logo about the line $y = -1$. Find the volume.

Part 2: Vase Engineering

Day 2 · Hollow vase: outer 4 + s, inner 3.5 + s, $s = \sin\left(\frac{\pi x}{10}\right)$, $0 \le x \le 20$.

5. Find the glass volume and the inside (water) volume.

6. The wall thickness is changed to w (inner r = 4 − w + s). Write the glass volume as a function of w.

7. The vase must hold at least 850 cm^3 of water. What is the greatest wall thickness allowed?

8. Find the glass cost for that thickness.

Part 3: Paperweights

Day 3 · Base under $y = 4 - x^2$, $y \geq 0$.

9. Compare the volumes with square and semicircular cross sections perpendicular to the x-axis.

10. Rotate the base about the y-axis. Use the disc method in y to find the volume.

11. Write the design portfolio with three products, their integrals, volumes, and prices.

Final Product: Product Design Portfolio

Your final product must include:

- Logo areas including split and rotations
- Vase thickness optimization
- Paperweight comparisons
- Prices for each product
- Design portfolio

Reflection

R1. How does rotating about a different axis change the radius?

R2. Why did the water-volume requirement limit wall thickness?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Areas and volumes are set up and evaluated correctly and used to price and design products. Goes beyond the minimum with a creative, realistic choice.	Areas and volumes are set up and evaluated correctly and used to price and design products.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Portfolio includes diagrams, integrals, results, and prices. Polished, organized, and visually engaging.	Portfolio includes diagrams, integrals, results, and prices. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.